

Pythagoras' Theorem & Trigonometry

Extra practice test 3 · 27 marks total

Name: _____

Date: _____

Score: _____ / 27

Section A — Pythagoras' theorem

A1. A right-angled triangle has shorter sides of 8 cm and 15 cm. Find the length of the hypotenuse.

[2 marks]

A2. A right-angled triangle has a hypotenuse of 25 cm and one shorter side of 7 cm. Find the length of the other shorter side.

[3 marks]

A3. A rectangle measures 20 cm by 21 cm. Find the length of its diagonal.

[2 marks]

Section B — SOH CAH TOA basics

B1. Which SOH CAH TOA ratio links the opposite and adjacent sides (not the hypotenuse)?

[1 mark]

B2. Write down the SOH CAH TOA formula you would use to find the hypotenuse if you know the opposite side and the angle.

[1 mark]

B3. Write down the SOH CAH TOA formula you would use to find an angle if you know the adjacent and hypotenuse.

[1 mark]

Section C — Finding missing sides

C1. A right-angled triangle has an angle of 48° and a hypotenuse of 14 cm. Find the length of the side opposite the angle, to 1 decimal place.

[3 marks]

C2. A right-angled triangle has an angle of 29° and an adjacent side of 11 cm. Find the length of the hypotenuse, to 1 decimal place.

[2 marks]

C3. A right-angled triangle has an angle of 51° and an adjacent side of 4 cm. Find the length of the side opposite the angle, to 1 decimal place.

[2 marks]

Section D — Finding missing angles

D1. A right-angled triangle has an opposite side of 5 cm and a hypotenuse of 12 cm. Find the angle, to 1 decimal place.

[2 marks]

D2. A right-angled triangle has an adjacent side of 8 cm and an opposite side of 6 cm. Find the angle, to 1 decimal place.

[3 marks]

Section E — Applying Pythagoras & trigonometry

E1. A ladder rests against a vertical wall. The foot of the ladder is 1.5 m from the wall and the ladder is 4 m long. Calculate how far up the wall the ladder reaches, to 1 decimal place.

[2 marks]

E2. A kite is flying on a taut string 25 m long, which makes an angle of 40° with the ground. Calculate the height of the kite above the ground, to 1 decimal place.

[3 marks]